

Scientific Forgery Image Detection

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1. Introduction

1.1 Purpose

This document defines the requirements for the Scientific Forgery Image Detection System. The system is designed to detect copy-move forgery in biomedical and scientific images, where regions of an image are duplicated to manipulate research results.

The system will allow users to upload scientific images through a web interface. These images will be processed by a backend system designed to support future detection functionality.

1.2 Intended Audience

This document is intended for:

- Software developers working on the system
- Course instructors evaluating the project
- Students involved in system design
- Future maintainers of the software

1.3 Overall Description

The system is a web-based application consisting of:

- A frontend user interface for uploading images
- A backend API for receiving and processing images
- A planned detection system for analyzing scientific images

At this stage, the system is in early development and primarily focuses on:

- Project setup
- Image upload functionality
- Backend communication
- Basic workflow structure

No advanced image detection model is implemented yet.

2. External Interface Requirements

2.1 User Interface Requirements

The system will provide a simple web interface with the following components:

- Homepage displaying project information
- Image upload page
- Upload button for selecting image files
- Submit button to send image to backend
- Basic result display area

The interface must be simple, accessible, and user-friendly.

2.2 Software Interface Requirements

The system will interact with the following software components:

Frontend:

- React.js for UI development
- HTML/CSS for layout and styling

Backend:

- Python as the programming language
- FastAPI for API development
- Uvicorn as the server runtime

Communication:

- HTTP requests between frontend and backend
- JSON responses from backend

2.3 Hardware Interface Requirements

No specialized hardware is required. The system will run on standard computing devices capable of running a web browser and local development servers.

2.4 Communication Requirements

- REST API communication between frontend and backend
- Image files transmitted via HTTP requests
- Responses returned in JSON format

3. Legal and Ethical Considerations

3.1 Data Integrity

The system processes scientific images, which may contain sensitive or research-related content. All uploaded images should be handled securely and not altered except for analysis purposes.

3.2 Privacy Considerations

- Uploaded images are used only for detection purposes
- No user personal data is required
- Data should not be shared externally without permission

3.3 Ethical Concerns

This system is designed to support scientific integrity by detecting possible image manipulation. However:

- Results should not be considered absolute proof of fraud
- The system should be used as a supporting tool
- Misinterpretation of results must be avoided

4. Glossary

Term	Definition
API	Interface used for communication between software components

Backend	Server-side system that processes requests
Frontend	User interface of the application
Copy-Move Forgery	Image manipulation technique where parts of an image are duplicated
FastAPI	Python framework for building APIs
React.js	JavaScript library for building user interfaces
HTTP	Protocol used for web communication
JSON	Lightweight data format used for API responses

5. Versions Table

Version	Date	Description	Authors
1.0	May 8, 2026	Initial SRS created for Snapshot 1	Janelle Rivera, Eric Marroquin, Alexis Flores

6. References

- FastAPI Documentation: <https://fastapi.tiangolo.com/>
- React Documentation: <https://react.dev/>
- OWASP API Security Guidelines: <https://owasp.org/www-project-api-security/>